Lily (Shi-Tao) Zhang

Cleveland, OH | zlily0049@gmail.com | lilyzhang4499-cmd.github.io

Professional Summary

Master of Science in Biostatistics with a Ph.D. in Biochemistry and Molecular Biology and over six years of biomedical research experience. Skilled in applying statistical methods to clinical and population health data; developing reproducible analytical workflows; evaluating model assumptions; and communicating results to multidisciplinary research teams. Proficient in R, SAS, and Python, with experience in regression modeling, survival analysis, causal inference, and large-scale healthcare datasets.

Technical Skills

Programming: R, SAS, Python
Statistical Methods: Generalized Linear Models, Survival Analysis, Causal Inference, Propensity Score Methods, Longitudinal and Cross-sectional Data Analysis
Data Management and Reporting: Data Cleaning, Data Wrangling, Missing Data Handling, Reproducible Workflows, Quarto **Healthcare Data:** NHANES, HCUP, BRFSS, HRS

Experience

Biostatistics Intern **May 2026–July 2026**
Case Western Reserve University, Cleveland, OH
Inflammatory Extracellular Vesicles Promote Immunosenescence and Frailty Risk in People Aging with HIV Infection

- Analyzed longitudinal plasma and extracellular vesicle biomarker data using R.
- Applied generalized linear models, time-to-event methods to identify biomarkers associated with frailty outcome and progression.
- Evaluated model assumptions, performed multiple-testing correction, and developed reproducible analysis pipelines, publication-quality figures, and statistical reports.
- Collaborated with PI and biomedical researchers to translate statistical findings into biologically meaningful conclusions.

Postdoctoral Fellow **March 2022–July 2025**
Case Western Reserve University, Cleveland, OH

- Designed and conducted computational and experimental studies of protein structure and function.
- Analyzed complex biological data, created scientific visualizations, and communicated findings through peer-reviewed publications and presentations.
- Collaborated with multidisciplinary researchers and contributed biological domain expertise to study design and interpretation.

Selected Projects

Comparative Effectiveness of Diabetes Medications — NHANES 2017–2020

- Used propensity score matching to compare metformin and sulfonylurea users and reduce treatment-selection bias.
- Built linear and logistic regression models for HbA1c-related outcomes and performed diagnostics and sensitivity analyses.

Prediction of Health Outcomes in Older Adults — HRS 2022

- Processed large-scale survey data, constructed analysis variables, and developed regression models for self-rated health and chronic conditions.
- Evaluated model performance and translated findings into clear summaries for nontechnical audiences.

Survival Analysis of Clinical Outcomes

- Built Cox proportional hazards models in R and SAS, assessed proportional hazards assumptions, and estimated hazard ratios with confidence intervals.
- Generated Kaplan–Meier curves and interpreted time-to-event results for biomedical research questions.

Education

M.S., Biostatistics, Case Western Reserve University, Cleveland, OH — Aug 2026
Ph.D., Biochemistry and Molecular Biology, Jilin University, China — Dec 2021
B.S., Life Science, Jilin University, China — June 2015

Publications and Certification

Zhang S. et al. *Frontiers in Chemistry*, 2021; Zhang S. et al. *Molecules*, 2021
Certification: SAS 9.4 Programming Fundamentals